My research experience // vision can be summed into three interlocking objectives, united under a goal of developing AI systems that understand, collaborate with, and empower the people who use them.

1. Understanding Human–AI Collaboration

Building capable and safe AI requires understanding how humans and models interact, not just how models perform alone. How do interaction dynamics influence joint efficiency, engagement, and *fulfillment*? How is cognitive load distributed between humans and models, and what mechanisms govern this process? Which aspects of interaction (communication clarity, initiative, calibration of uncertainty) enable coordination, especially as interaction becomes *multimodal* through language, voice, and vision? By examining these processes, we not only learn how to design better collaborators but also how to prevent failure modes in deployment: when models, for instance, learn to manipulate human preferences through sycophantic outputs.

In pursuit of this mission, I have led several lines of work: (1) In ***"When Models Know…"*** [1], I led a study with over 110 participants to develop a framework that systematically measures **how effectively models communicate knowledge to humans**, showing that a model's solo performance does not necessarily predict its ability to co-work effectively with people. We uncover distinct interaction profiles that govern how knowledge is transferred between humans and AI systems. (2) In the ***USACO*** [2, 5] project, we examined **human-AI collaboration in competitive programming** settings, finding that even non-expert human input, limited to algorithmic guidance, can significantly influence the performance/strategy of weak models. (3) At *Sierra*, I led efforts on **post-training for voice agents** [blog post]. Even with unresolved issues like imperfect ASR and unnatural TTS, voice interaction already shows how multimodality can shape interaction. For example, we find that post-training behaviors that seem minor in text (e.g., conciseness/repetitiveness) can meaningfully alter how users engage with voice agents, increasing satisfaction, turn-taking, and even intent divergence.

2. Scalable Training/Evaluation of Human-AI Collaboration

As we uncover the dynamics of collaboration, the next step is to translate these insights into mechanisms for evaluating and training AI systems. But while human evaluation remains essential, it is difficult to reproduce and scale, introducing *significant* noise and cost. This limitation has driven much public effort toward autonomous benchmarks: not because human judgment is unimportant, but because publicly aligned progress depends on *reproducibility*: datasets, code, and metrics that can be shared and replicated at scale. The result is a world in which only large corporations possess the resources to align AI with human values, often in ways that **reflect *private incentives* rather than public good** and with *significant lack of regulation*. To this end, I am **broadly interested in scalable training, evaluation, and alignment of human-ai interaction**. One promising path involves building computational models that can simulate human interaction and values // and constructing realistic environments that reward models for effectively collaborating with these computational proxies.

My past work has explored two complementary approaches to this. (1) At Sierra, I lead the third iteration of **Tau-Bench** [3], one of the first widely adopted benchmarks that uses steerable, prompt-based trait descriptions to simulate users in goal-oriented collaboration. Our new approach uses a structured-to-unstructured data generation pipeline to programmatically construct difficult, ambiguous, yet realistic collaboration scenarios. We are additionally extending evaluation across modalities, including voice, enabling one of the **first full-duplex, task-oriented, voice-agent benchmarks**. (2) In contrast, **IMPersona** [4] trains models on **up to ten years of an individual's text data** to produce AI impersonations of that person: a step toward *policy-based simulation*, where we learn to reproduce an individual's behavioral policy directly from their responses across situations rather than from natural language trait descriptions. I measured success via a 120+ person study involving an *individual-level* Turing test, evaluating whether close acquaintances could distinguish the real person from their AI

impersonation. Our method led fine-tuned llama-8b to be misidentified as the real person 44% of the time, while GPT-4o with few-shot prompting fooled participants only ~5% of the time.

IMPersona, in a way, sacrifices steerability for realness: just as in real life, human behavior cannot be neatly defined by static traits like “angry” or “empathetic,” and are extremely context dependent. LLMs, trained on the behaviors of countless individuals, already internalize a latent structure of *real* human behavior; what they lack is alignment to any *single* individual’s consistent policy. Some of my ongoing work aims to make this structure explicit: uncovering recurring behavioral patterns and **mapping them to semantic dimensions**. A related challenge is **modeling *asymmetric knowledge profiles***: teaching models not only to simulate personalities but also cognitive limitations. Not only are these problems so difficult, but they grasp at the fundamentals of what makes us human, which is deeply exciting.

3. Generational impact of AI on Humans:
We often discuss adapting AI to collaborate with humans as if “humans” were static, but we continually evolve alongside our tools. Each major shift in technology reshapes how we think: for example, one could argue search engines turned us from *memorizers* into *retrievers*, people who instead of memorizing knowledge, memorize how to find it. Large language models extend that trajectory: they let us offload reasoning itself. We are moving from *problem-solvers* to *orchestrators*, guiding and critiquing reasoning across a human-AI collective.

As this mode of thinking becomes native to a new generation, can we study these shifts to guide education, policy, and design toward healthier and more empowering human-AI relationships? To this end, I collaborate with teachers in my hometown in West Virginia to integrate AI tools into classrooms: some current work involves **evaluating AI capability to produce worksheets that elicit engagement**. I hope to formalize these efforts into longitudinal studies that trace how students’ learning patterns adapt alongside advancing AI systems, especially in small-town and underrepresented communities.

Conclusion: The beauty of it all is that as humans change and adapt to the technologies they engage with, so too do the modes of collaboration we study in Goal 1. These evolving interactions, in turn, inform new computational models of human behavior. And so the feedback loop continues: a never-ending co-adaptation between model and human, each learning from the other, continually advancing toward the shared goal of enabling humans to achieve their fullest potential.

References:

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